Module 9

Explaining Network Security Concepts

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Learning Objectives

- Explain common security concepts.
- Distinguish risk, vulnerability, exploit, and threat.
- Explain the importance of audits and regulatory compliance.
- Summarize types of attacks and their impact on the network.

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Lesson 9.1

Security Concepts

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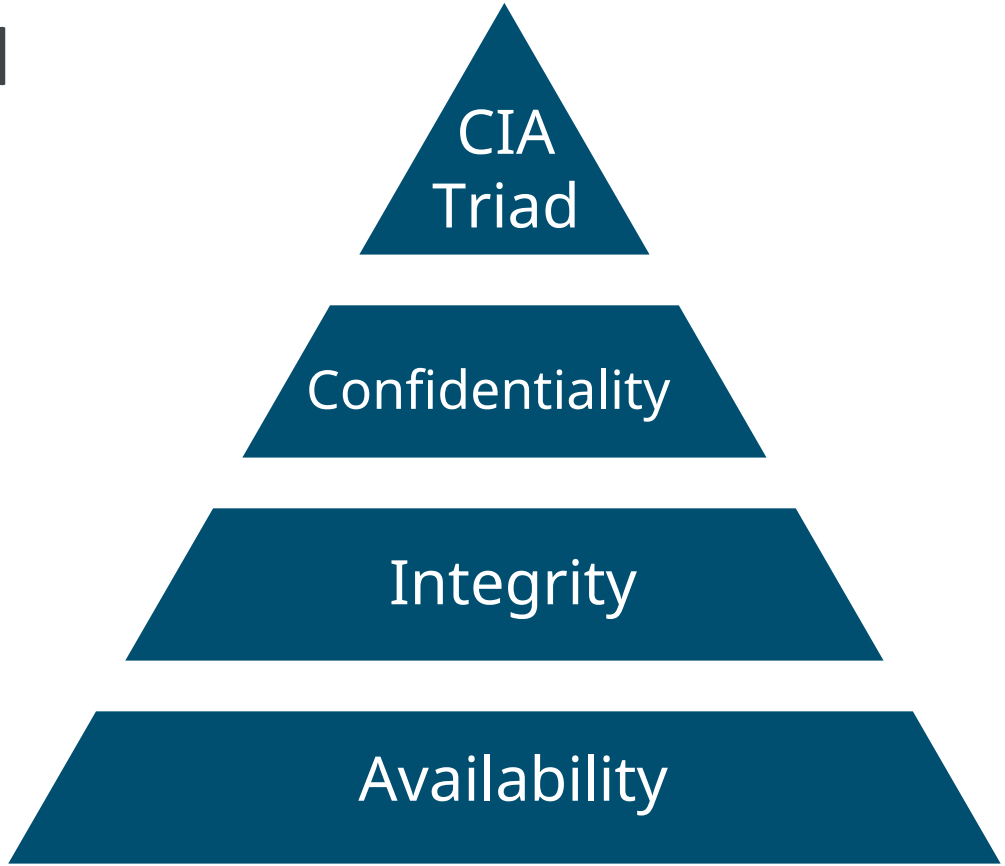
CIA Triad



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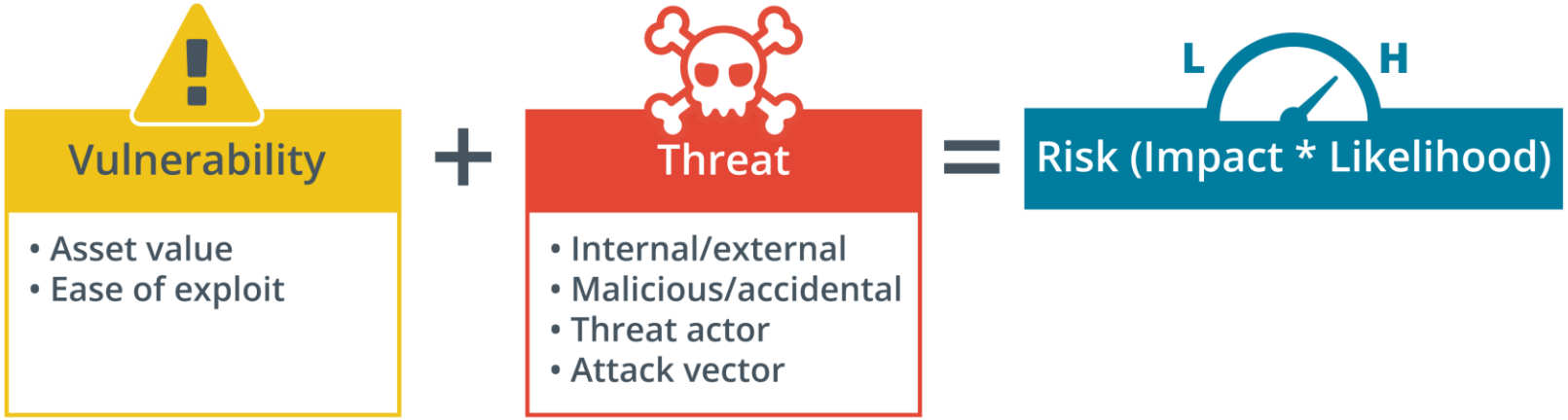
Vulnerability, Threat, Risk



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Security Audits

Security Audit

- Systematic evaluation of a company's information system security
- Measure conformity to established security criteria
- Identifies strengths and weaknesses
- Highlights areas for security improvement

Key Components

- Policies assessment
- Procedures review
- Technical controls evaluation
- Access controls inspection
- Risk management practices assessment

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Types of Security Assessments

Compliance Audits

- Verify adherence to laws, regulations, and standards

Risk-Based Audits

- Identifying and prioritizing potential threats/vulnerabilities

Technical Audits

- Deep-dive into IT infrastructure (e.g., network security, access controls, and data protection)

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Role of Risk Management

Identifying Critical Assets

- Determine mission essential functions/assets vital to business operations

Business Impact Analysis (BIA)

- Assess potential losses from threat scenarios (e.g., DoS attack)

Mitigation Strategies

- Balance cost of security controls against the potential risks to determine acceptable risk levels

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Regulatory Compliance

Definition

- The process by which organizations ensure they are following all relevant laws, regulations, and guidelines applicable to their industry

Purpose

- To protect data, individuals' privacy, and ensure the integrity of financial transactions and sensitive information

Key Components

- Implementation of security measures and controls
- Regular internal and external audits
- Compliance with specific industry standards

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Regulatory Compliance

General Data Protection Regulation (GDPR)

- Scope: collection/ processing of personal data in the EU
- Informed consent and data rights

Payment Card Industry Data Security Standard (PCI DSS)

- Scope: organizations handling credit card transactions
- Standards for protecting cardholder data

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Understanding Encryption

What is Encryption?

- Converting human-readable data (plaintext) into a coded format (ciphertext)
- Only accessible to authorized users with decryption key

Purpose of Encryption

- Ensures data confidentiality
- Protects information from unauthorized access

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Types of Cryptographic Algorithms

Encryption Algorithm

- Converts plaintext into ciphertext (key required for decryption)
- Use Case: Protecting emails or files

Cryptographic Hash Algorithm

- Transforms a string into a fixed-length hash (cannot be reversed)
- Use Case: Password storage, verifying data integrity

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The States of Data

Data at Rest

- Persistent storage media protection
- Example: encrypted hard drives

Data in Transit

- Protection while data is transmitted
- Example: TLS encryption

Data in Use

- Securing data in volatile memory
- Example: RAM encryption techniques

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Vulnerabilities

Definition	Causes	Impact	Targets
• Flaws in software design allowing bypass of security or causing crashes	• Misconfigurations, poor practices, design faults	• Allows attackers to execute arbitrary code, install malware, compromise security configurations	• Commonly include web servers, browsers, plug-ins, email clients, databases

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Types of Exploits and Vulnerabilities

Zero-Day Vulnerabilities

- Exploited before developers can patch
- High destructive potential

Unpatched/Legacy Systems

- Pose significant threats due to lack of updates or support

Vulnerability Assessment

- Evaluating a system's security based on its configuration
- Verify it matches the ideal baseline
- Involves manual inspections and automated scans

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Deception Technologies

What are Deception Technologies?	Key Components	Benefits	Placement
• Confuse and divert cyber attackers • Make attacks more difficult and costly	• Honeypots: Fake systems to lure attackers • Honeynets: Fake networks for broader trapping	• Early attack detection via decoy engagement • Analysis of attacker tactics for better defense • Early warning by spotting breaches early	• Set in monitored areas between Internet and internal network

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Activity: Quick Search

Run a quick search for
“zero-day attacks that happened in the last week.”

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Lesson 9.2

Network Threats and Attacks

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External Threats

Origin

Attacks or malicious activities by individuals/ groups from outside the organization

Examples

- Hackers
- Cyber Criminals
- Espionage
- Competitive intelligence

Characteristics

- Often sophisticated
- Targeted
- Relentless

Mitigation Strategies

- Firewall protection
- Intrusion detection systems (IDS)
- External audits
- Security awareness training

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Internal Threats

Origin

- Originate from within the organization
- Often involve employees, contractors, or business partners

Examples

- Accidental data leaks
- Deliberate data theft
- Sabotage
- Insider trading

Characteristics

- Harder to detect due to legitimate access
- May stem from dissatisfaction, malicious intent, or carelessness

Mitigation Strategies

- Role-based access controls
- User activity monitoring
- Regular security training
- Clear policies on data handling

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Enumeration Attacks

Footprinting

- Network topology discovery
- Security configurations via social engineering or port scanning

Fingerprinting

- Identifying system/software versions
- Vulnerability probing

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Denial of Service (DOS) Attacks

Origin

- Internet's inception and growing more complex
- Evolved to sophisticated DDoS using botnets

Characteristics

- Intent: Disrupts service, denying user access.
- Methods: Uses resource exhaustion, bandwidth saturation, software exploitation
- Impact: Service degradation to complete shutdown

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DDoS Attacks – Examples and Mitigation

Examples of DDoS Attacks

- ICMP Flood: Overloads with ICMP packets (unreachable targets)
- SYN Flood: Abuses TCP handshake (blocks legitimate server access)
- DNS Amplification: Exploits misconfigured DNS servers (attack traffic)

Mitigation Strategies

- Early Detection: Monitors for unusual traffic patterns
- Traffic Filtering: Employs firewalls and IDS to block malicious traffic
- Response Plan: Prepares specific actions for DoS attacks
- Redundancy: Implements network redundancy to reduce downtime

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Botnet Attacks

Definition

- Network of compromised computers for malicious use
- Infected by malware and controlled remotely

Types of Botnets

- Distributed Denial of Service (DDoS)
- Spam botnets
- Banking trojan botnets

Mitigation strategies

- Implement cybersecurity like firewalls and antivirus
- Analyze network for unusual traffic
- Isolate and fix compromised devices
- Teach safe internet use

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Malware

Definition

- Harmful software disrupting or damaging systems

Types of Malware

- Viruses/Worms
- Trojan
- PUPs/PUAs

Vectors and Payloads

- Vectors: Infection and spread methods
- Payloads: Executed harmful actions (e.g., spying, unauthorized access, data encryption for ransom)

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Fileless Malware

Characteristics

- Sophisticated threat, no code on disk, hard to detect

Execution Methods

- Resides in system memory, uses scripting tools (e.g., PowerShell, WMI)
- Persistence via registry changes, no file-based methods

Detection Challenges

- "Live off the land" techniques, shellcode for stealth
- Part of APTs and AVTs, low observable characteristics, advanced evasion

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Activity: 5 Second Game

You have 5 seconds to:

List 5 Types of Cyber Attacks

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Spoofing Attacks

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Spoofing Attacks

Definition

- Disguising oneself as someone else to gain unauthorized access

Purpose

- Trick users/devices, bypass security, and steal data or spread malware

Types

- IP spoofing
- ARP spoofing
- Email spoofing

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On-Path Attacks

Definition

- Attacks that intercept and possibly alter two parties' communications undetected

Purpose

- Steal sensitive personal or corporate information
- Inject malware

Common Types

- Session hijacking
- SSL stripping
- DNS spoofing
- Wi-Fi eavesdropping
- ARP spoofing

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ARP Spoofing Example

No.	Time	Source	Destination	Protocol	Length	Info
6	10.022521400	Microsof_01:ca:4a	Microsof_01:ca:76	ARP	42	10.1.0.102 is at 00:15:5d:01
7	10.032593900	Microsof_01:ca:4a	Microsof_01:ca:77	ARP	42	10.1.0.2 is at 00:15:5d:01:c
8	10.032605300	Microsof_01:ca:4a	Microsof_01:ca:76	ARP	42	10.1.0.101 is at 00:15:5d:01
9	18.219200600	10.1.0.101	10.1.0.2	TCP	66	1702 → 80 [SYN] Seq=0 Win=65
10	18.220473400	10.1.0.101	10.1.0.2	TCP	66	[TCP Out-Of-Order] 1702 → 80
11	18.223616200	10.1.0.2	10.1.0.101	TCP	66	80 → 1702 [SYN, ACK] Seq=0 A
12	18.228456800	10.1.0.2	10.1.0.101	TCP	66	[TCP Retransmission] 80 → 17
13	18.228797700	10.1.0.101	10.1.0.2	TCP	54	1702 → 80 [ACK] Seq=1 Ack=1
14	18.229264100	10.1.0.101	10.1.0.2	HTTP	433	GET / HTTP/1.1
15	18.238162600	10.1.0.101	10.1.0.2	TCP	54	1702 → 80 [ACK] Seq=1 Ack=1
16	18.238250400	10.1.0.101	10.1.0.2	TCP	433	[TCP Retransmission] 1702 →
17	18.239342200	10.1.0.2	10.1.0.101	HTTP	412	HTTP/1.1 302 Redirect (text
18	18.244530700	10.1.0.2	10.1.0.101	TCP	412	[TCP Retransmission] 80 → 17
19	18.245021200	10.1.0.101	10.1.0.2	TCP	54	1702 → 80 [ACK] Seq=380 Ack=
20	18.252481800	10.1.0.101	10.1.0.2	TCP	54	[TCP Dup ACK 19#1] 1702 → 80
21	18.255190400	10.1.0.101	10.1.0.2	TCP	66	1703 → 443 [SYN] Seq=0 Win=6
22	18.260503200	10.1.0.101	10.1.0.2	TCP	66	[TCP Retransmission] 1703 →
23	18.261065300	10.1.0.2	10.1.0.101	TCP	66	443 → 1703 [SYN, ACK] Seq=0
24	18.268454300	10.1.0.2	10.1.0.101	TCP	66	[TCP Retransmission] 443 → 1

```
▸ Frame 9: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
▾ Ethernet II, Src: Microsof_01:ca:77 (00:15:5d:01:ca:77), Dst: Microsof_01:ca:4a (00:15:5d:01:ca:4a)
    ▸ Destination: Microsof_01:ca:4a (00:15:5d:01:ca:4a)
    ▸ Source: Microsof_01:ca:77 (00:15:5d:01:ca:77)
      Type: IPv4 (0x0800)
```

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MAC Flooding Attacks

Definition

- Overloads switch CAM table with many MAC addresses, causing switch failure and traffic broadcast

Purpose

- Disrupt switch function to eavesdrop on normally inaccessible network traffic

Types of MAC Flooding Attacks

- Random MAC address flooding
- Incremental MAC address flooding
- Targeted MAC flooding

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VLAN Hopping Attack

Definition

- Exploit where attackers send packets to a VLAN without access, using vulnerabilities in VLAN implementation on switches

Purpose

- Bypass security, accessing restricted/sensitive networks

Types

- Switch Spoofing
- Double Tagging

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Activity: Fill in the Blank

1. Two types of _____ hopping attacks include switch spoofing and double tagging.
2. A/an _____ attack involves intercepting and possibly altering two parties' communications without detection.
3. A/an _____ flooding attack overloads switch CAM table with many addresses, causing switch failure and traffic broadcast

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Rogue System Attacks

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Rogue Devices and Services

Definition

- Unauthorized hardware and software that connect to a network without permission, potentially causing serious security risks

Characteristics

- Unauthorized access
- Malicious intent
- Data interceptions and manipulation

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Rogue DHCP Server Attacks

Definition

- An unauthorized DHCP server distributing incorrect network configuration parameters to clients

Characteristics

- Distribution of invalid configuration
- Traffic redirection
- On-path attacks

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DNS Attacks

Definition

- Attacks that target the DNS, undermining the integrity and availability of internet services

Characteristics

- Disruption of service
- Domain hijacking
- Cache poisoning

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Activity: Two Truths and a Lie

Rogue devices and services are unauthorized hardware and software that connect to a network without permission, potentially causing serious security risks

Rogue DNS Server Attacks involve an unauthorized server distributing incorrect network addresses to clients

DNS Attacks target the system which translates domain names into IP addresses, undermining the integrity and availability of internet services

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Lesson 9.5

Social Engineering

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Social Engineering Attacks

Definition

- Manipulative technique cyber criminals use to exploit human vulnerabilities, not technical weaknesses

Goal

- Deceive individuals into giving up confidential or personal information

Characteristics

- Highly effective due to exploiting people's natural tendency to trust
- Success relies on the attacker appearing trustworthy or authoritative
- Persuades victims to breach security practices or ignore red flags

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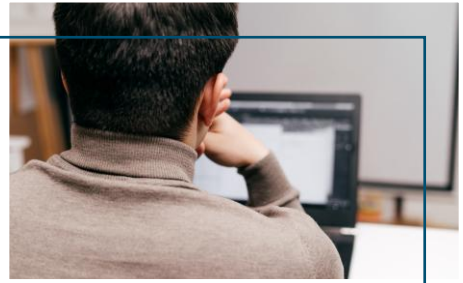
Types of Social Engineering Attacks



Shoulder Surfing

Dumpster Diving

Tailgating/Piggybacking

Phishing

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Password Attacks

Definition	Dictionary Attacks	Brute Force Attacks
• Attempts to obtain or bypass individuals' passwords using various methods	• Enters all dictionary words • Targets weak, simple passwords	• Tries all character combinations • Time-consuming but cracks any password

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Summary

- **Policies & Controls**: Establish policies and deploy controls aligned with the CIA triad (Confidentiality, Integrity, Availability)
- **Assessment & Monitoring**: Use tools and processes to continuously evaluate vulnerabilities, threats, and risk
- **Awareness Training**: Educate users on common attacks like footprinting, spoofing, DoS, DNS manipulation, VLAN hopping, malware, password cracking, and social engineering

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